

Hazard Traceability Matrix

Product:

DocID:

Participants:

Issued by:

Reviewed by:

Appr. by:

Project:

Rev:

Date: _____

Date: _____

Date: _____

Sign: _____

Sign: _____

	Risk analysis							Risk eval.	Risk control										
ID	Hazard	Reasonably foreseeable sequence or combination of events	Hazardous Situation	Harm	Notes	Po	Severity	Acceptable?	Risk control options and rationale	Risk control measure	Risk control verification	Risk control verification records	Impl.?	R-Po	R-Severity	Has the risk been reduced AFAP?	Is the benefit greater than the residual risk?	Residual risk	Comment on residual risk / Benefit-risk analysis
Example risks for a device that comes in contact with the patient's skin without being invasive																			
	Improper data normalization	The unusually high values in a variable was not normalized properly, leading to an inaccurate cluster, dominated by a variable	Inaccurate prediction of TBI severity	Delayed care		2	4												
	Missing biomarker data	A blood draw on the 48h mark was missed or failed. The model makes clusters based on incomplete data or ignores the patient	Inaccurate prediction of TBI severity	Delayed care		3	4												
	Delayed syncing of data to the MATLAB GUI	The 24h or 48h blood sample is analyzed, but network lag occurs. The prognosis is made based on the 0h mark	Inaccurate prediction of TBI severity	Delayed care		3	5												
	Lack of model interpretability (black box)	The model assigns the patient to a certain cluster. The clinician could attributes the risk to an incorrect biomarker	Misinterpretation of biological cause of severity score	Inappropriate care		4	3												
	Cluster overlap	A patient's biomarker falls under the moderate and severe categories and the model assigns them to moderate	The clinician fails to indicate that the patient is borderline severe	Delayed care		4	4												

	Biomarker concentration outliers	A patient has an unusually high concentration of a certain biomarker, which causes the model to shift the centroid from severe to moderate (or vice versa)	Misclassification of patient, receiving FNs and FPs	Inappropriate care		3	4											
	Identification of non-clinical noise as a distinct phenotype	The model is forced to find K clusters in a high variance data. Then it creates a new phenotype and classified as unique risk	Misinterpretation of data fluctuation	Inappropriate care		4	3											

